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Decoding Contractual Architectures: Industrial Practices in Agricultural Data Governance Reveal Systemic Derivative Inequities

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Abstract

The exponential growth of agricultural data, driven by smart sensors and AI, has revolutionized farm data management but intensified governance challenges regarding ownership, control, and value distribution. This study bridges a critical empirical gap by analyzing contractual documents from U.S.-based Agriculture Technology Providers (ATPs) and policy frameworks, employing NLP techniques and knowledge graph modeling to map data governance logics. Findings reveal ATPs leverage intellectual property and trade secret specifications to obtain control over data derivatives, such as analytical insights and predictive models, while acknowledging farmer ownership of raw data. Despite voluntary industry initiatives affirming ethical principles (e.g., farmer data ownership), non-binding terms and opaque contractual language indicate power asymmetries, enabling ATPs to unilaterally monetize derivatives without equitable profit-sharing. Crucially, ATPs' governance stances evolve from open data in startup phases to aggressive restrictions upon maturity, exacerbating inequities and stifling innovation. The research underscores an urgent need for regulatory interventions that redefine ownership to include co-ownership of derivatives, enforce transparency, and implement traceability mechanisms to audit data flows. These reforms are essential to foster an equitable, sustainable digital agriculture ecosystem.

1 Introduction

With millions of sensors installed for agricultural purposes, the current average data points collected by a farm are estimated to be approximately half a million data points per day (Meola, 2021) and are projected to reach 4 million by 2036 (Medori, 2019). A "Big" amount of data will be harvested on each farm to describe every aspect of its farming process, thereby paving the way for the sustainability of future agricultural systems (Basso & Antle, 2020). By leveraging smart sensors and Artificial Intelligence (AI) algorithms, insights from "Big" data collected on farms can be interpreted to direct farming operations and behaviors for higher efficiency (Eastwood et al., 2019; Wolfert et al., 2017). The implementation of data-oriented technologies represents a revolutionary shift, transforming the relationships among stakeholders, including farmers and Agriculture Technology Providers (ATPs), from their traditional roles (Jakku et al., 2019; Wolfert et al., 2017). Concerns about how agricultural data is governed, particularly who owns, accesses, and controls it, have grown substantially (Bronson & Knezevic, 2016).

Data governance refers to the policies, standards, and practices that regulate the collection, storage, use, and sharing of data. In agriculture, this governance is complicated by the diverse actors involved, including farmers, agribusinesses, ATPs, and data aggregators. Unlike sectors such as health or finance, agriculture lacks standardized frameworks or regulatory structures for governing data rights, leading to uncertainty and asymmetry in how data is managed and shared (Carbonell, 2016). Yu et al. (2025) revealed that ATPs' data governance perspectives evolve from open data to conservative control as they develop from startups to mature companies.

Current efforts and investigations into data governance are generally focusing on personal data and reflect a complex and evolving landscape, driven by legal, ethical, and social challenges. Scholars and policymakers are increasingly debating the conceptual and practical implications of treating data as property. Hummel et al. (2021) argue that data ownership claims can be framed around four key dimensions: property rights, socio-cultural recognition, redistributive justice, and participation. Hicks (2023) underscores the importance of moving beyond regulatory frameworks by exploring data commons and cooperative models that introduce collective governance and social interactions.

Emerging solutions, such as data trusts and data cooperatives, offer alternative governance models to balance individual control and communal benefits (Lanier, 2014; Thouvenin et al., 2017). These solutions seek to address concerns raised by the potential monopolization of data resources, which could restrict innovation and undermine societal interests (Purtova, 2015; Rodwin, 2009). Moreover, Pearce (2018) advocate for hybrid models that blend property and contractual rights to protect both personal data and public interest. The ongoing discourse suggests that a multifaceted approach involving legal reforms, participatory governance, and ethical reflection is essential to navigate the intricate landscape of data ownership in the digital age. However, the ambiguity of data as a non-rivalrous and intangible resource presents significant legal hurdles (Hummel et al., 2021). Traditional property laws, particularly in jurisdictions like the U.S. and EU, struggle to reconcile data rights with existing property concepts, often defaulting to privacy protection rather than full ownership frameworks (Contreras et al., 2018; Purtova, 2015). But agriculture data has few personal privacies and are mostly fact recording of natural phenomenon that cannot well fit the property law and privacy protection mechanism in the current legislation system.

Farmers often express concerns about privacy, loss of control, and potential misuse of data, particularly when it is transmitted to or processed by third parties (Maru et al., 2018). A study made by American Farm Bureau Federation (AFBF) in 2014 shows that farmers in the US are worried about the security, access, and use of their data. Wiseman et al. (2019) found that the majority of farmers in Australia are concerned about the control, ownership, and misuse of data and looking for clearer data agreements, greater governance, and trust. Although these concerns make farmers reluctant to adopt new technologies (Wiseman et al., 2019), the deep understanding of data governance require a thorough consideration of the entire value chain of data.

ATPs often prioritize protecting their Intellectual Property (IP) and proprietary systems in data governance decisions. Data collected through their platforms, such as machine logs or algorithm outputs, can reveal commercially sensitive information and are therefore closely guarded (Carbonell, 2016; Wiseman et al., 2019). Yu et al. (2025) revealed that ATPs' data governance perspectives evolve from flexible terms to aggressive restrictions across development stages and complicating business structures.

Despite the increasing attention to agricultural data governance, most existing research has focused on high-level principles such as ownership, privacy, trust, and consent. Studies on social topics are primarily

focusing on stakeholders' perspectives (farmers and ATPs). However, there is limited empirical analysis of how these principles are operationalized in practice. To date, no study has systematically examined the contractual conditions that define data rights and responsibilities between farmers and ATPs. This gap is significant, as contracts often serve as the primary mechanism through which data governance is implemented on the ground. Understanding these terms is essential to evaluating whether current governance practices are equitable, enforceable, and aligned with broader agricultural sustainability goals.

2 Methodology

The scope of this study is based on the stakeholders in the US. Thus, the data and information collected for investigation are all based on the US legal system and agriculture industry. Although the scope is narrowed and specific, the knowledge and vision presented by this study are insightful for general agriculture data governance that can be referred by other markets or countries.

2.1 Define data governance in agriculture

Data governance in agriculture should be understood as a set of legal and ethical frameworks that guide how data activities are conducted and how associated rights are defined and protected. These frameworks intersect with key legal concepts such as IP, ownership, and are shaped by the diverse perspectives of stakeholders including farmers, ATPs, and government entities. In this study, we review the legal literature on data ownership in agriculture, focusing primarily on the US, and summarize the key concepts and visions emerging from existing principles developed by industry associations and government agencies. This synthesis serves as a foundation for constructing a conceptual map of current data governance practices. In parallel, we incorporate stakeholder perspectives, including those of farmers and ATPs. Farmers' concerns and structural disadvantages, identified in previous research, are examined and compared with ATPs' positions, as revealed in an earlier study conducted by the authors. This comparative analysis supports a broader discussion on the future of data governance in agriculture. The resulting insights will be used to enrich the conceptual landscape by identifying emerging ethical trends and forecasting potential developments in governance models.

2.2 Data collection

This study analyzes data governance terms contained in contractual documents from ATPs and policy documents from selected government agencies. The methodology followed three sequential steps:

1. **ATP Identification**: A comprehensive list of relevant ATPs was identified and compiled using public data sources, including databases from Agritech Tomorrow, iGrow News, and USDA. All selected entities are either U.S.-based companies or international firms with business operations covering the U.S. agricultural market.

2. **Document Sourcing**: While formal contracts between ATPs and farmers/collaborators are typically confidential, partial governance terms are publicly disclosed through data policies, privacy policies, users' manuals, and other website materials. These documents, in some degree, reflect ATPs' actual practices.

3. **Data Collection**: Using Python's BeautifulSoup library, we developed a web scraper to extract relevant documents based on keywords like "privacy," "terms," and "user agreement." Extracted content was saved in plain text and filtered to remove boilerplate, irrelevant, and duplicate text, preserving only policy-relevant sections for analysis.

4. **Government Document Inclusion**: Additional documents from key government agencies (USDA, NSF, DOE, DOD) were obtained directly through their official websites.

2.3 Knowledge extraction

To analyze data governance practices across entities (ATPs, government agencies), we employ Natural Language Processing (NLP) techniques to extract two core elements identified in Section 3.1, data ownership and data utilization, from collected contractual documents. This structured methodology automates concept labeling of governance clauses (e.g., ownership assignment, commercialization rights), maps semantic relationships and categorizes logic patterns by entity type, transforming unstructured text into analyzable frameworks that embody the legal relationships within source materials. The extracted logic will be maps to legally attributed property defined in contractual documents. This logic chain (ownership → utilization → property) forms a foundational framework that categorizes and explains complex data governance concepts, including rights allocation, value capture, and derivative control, across ATPs. The specific relationships are transformed into custom subject–predicate–object triplets in a Neo4j graph database, enabling queryable, evolving representations of agricultural data rights relationships. Figure 1 demonstrates a conceptual sample of a collection of extracted relationships. It illustrates the flow and transformation of governing relationships among key stakeholders, Farmers, ATPs, and Third Parties, with “Data” as the central object of governance. Arrows represent semantically tagged relationships such as “provides data to”, “shares with”, “grants access to”. For example, a clause stating "Farmer grants ATP permission to share aggregated data with third parties" would be extracted and represented as:

Farmer → GrantsSharingRightsTo → ATP

ATP → SharesWith → Third_P

This kind of representation allows us to structurally analyze how data control and obligations flow through stakeholder networks, which forms the basis for sequential actions.

2.4 Mapping industrial practices

Data governance in agriculture is closely tied to the type of technology or service being used on the farm. The nature and sensitivity of the data collected vary significantly depending on the equipment, influencing how stakeholders approach data governance. For instance, data generated by a smart tractor may include detailed records of farming operations, such as cropping strategies, soil conditions, and yield estimates. This information reflects the farmer's operational knowledge and may be considered proprietary or commercially sensitive. Simultaneously, the same dataset might include machine operation logs, which can reveal insights into the proprietary control systems developed by the ATP. Disclosure of such information could pose risks related to IP protection or enable reverse engineering. Consequently, ATPs often take conservative terms on data governance to safeguard their technologies. In contrast, data from smart irrigation systems typically focus on environmental variables—such as soil moisture, temperature, and local weather conditions. These data points, while valuable for operational decisions, are generally less tied to proprietary farming strategies or ATP-developed systems. As such, they pose lower competitive risks if shared or disclosed. This variation illustrates that data governance in agriculture cannot be approached as a one-size-fits-all solution.

To capture a comprehensive information about data governance practices, various types of services provided by ATPs are categorized in Table 1, showing the distribution of 150 + selected ATPs across six key service categories, each contributing uniquely to data generation in modern agriculture. Remote Sensing and Analytics leads with 34 ATPs, reflecting strong industry investment in technologies that collect spatial, spectral, and temporal data via satellites, drones, and field sensors. Machinery Manufacturers (28 ATPs), such as John Deere, contribute data from farm equipment operations, including GPS-based movement, machine usage, fuel consumption, and yield mapping. Seed and Genetic Technologies (26 ATPs) generate valuable biological and performance data, such as germination rates, phenotypic traits, and genetic markers. Agricultural Testing and Diagnostics (26 ATPs) focus on laboratory and in-field data related to soil health, plant disease, and nutrient levels. Crop Protection (22 ATPs) contribute pest, disease, and chemical application data, essential for integrated pest management strategies. Finally, Fertilizer and Soil Input Technologies (20 ATPs) capture data on soil composition, input efficiency, and nutrient application rates. These categories illustrate the diverse sources of agricultural data and underscore the complexity of managing data rights and interoperability across sectors in the digital agriculture ecosystem.

Table 1
Categories of services provided by ATPs

Service Category	Number of ATPs
Machinery Manufacturers	28
Remote Sensing and Analytics	34
Seed and Genetic Technologies	26
Crop protection	22
Fertilizer and Soil Input Technologies	20
Agricultural Testing and Diagnostics	26

3 Results

3.1 Legal frameworks and industrial perspectives

3.1.1 Legal frameworks

ATPs design data governance terms primarily to protect market competitiveness through IP frameworks, positioning data as proprietary assets. This approach faces fundamental legal challenges in the U.S., where there is no law specifying data as a legal property in that it lacks tangible attributes for classification as physical property (e.g., land or equipment) while failing to meet IP criteria since most farm data constitutes recordings of natural phenomena rather than creative output (Ellixson et al., 2019; Ferrell, 2014). Analogous to concert recordings capturing but not creating musical copyrights, farmers' data, whether soil metrics or yield logs, merely documents existing conditions without original intellectual contribution, rendering it non-copyrightable. Consequently, downstream ATPs exploited this void by freely utilizing farm data absent IP restrictions.

Scholars have explored trade secret protection as an alternative pathway (Ellixson & Griffin, 2016; Ellixson et al., 2019; Ferrell, 2014), but contractual complexities undermine this approach. For example, the End Users' License Agreement (EULA) by Climate LLC (2016), says:

By uploading, inputting, transmitting, storing or otherwise making Data available to the Climate Products, you agree that we may use, display, perform, reproduce, modify and distribute such Data without any compensation paid to you.

The obligations and responsibilities of Climate LLC regarding data from farmers are clearly described in the statement which is very specific to their partnership. Although raw data are usually non-copyrightable; trade secrets could be applied to raw data with two conditions: 1) controlled disclosure to keep the secrecy and 2) conferred economic benefits. Unfortunately, when referring to the statement in this agreement, farmers "*agree that we (Climate LLC) may use, … distribute such data*" indicating

relinquishing the control of disclosure of data. The economic benefits also cannot be represented since Climate LLC may use the data "*without any compensation paid to*" farmers. Therefore, the agreement rules out the possibility that the raw data can be considered as an IP in the form of a trade secret belonging to farmers.

With legal frameworks offering no clear ownership resolution, contracts become the practical governance mechanism, inherently favor ATPs (Wiseman et al., 2019). ATPs leverage legal expertise to draft agreements that bypass data's legal ambiguities while imposing unilateral terms. Farmers, lacking equivalent resources and bargaining power, face inherent disadvantages: even when understanding contractual implications (e.g., losing derivative rights), they possess no leverage to negotiate equitable terms. The Climate LLC case illustrates this power asymmetry, contracts become tools to extract value from legal gray areas rather than instruments of fair governance.

3.1.2 Industrial perspectives

Growing industry recognition of ethical imperatives has spurred voluntary initiatives, such as the U.S. *Privacy and Security Principles for Farm Data* (American Farm Bureau Federation, 2016), the EU *Code of Conduct* (Copa-Cogeca et al., 2018), and the Switzerland's *Charter on Digitalisation* (Agridigital, 2018), etc., which explicitly affirm farmer ownership of farm-sourced data and commit participating ATPs to embed these principles in contracts. However, as non-binding agreements lacking legal enforceability, they impose no obligations on ATPs; violations (e.g., unauthorized data commercialization) provide farmers no legal recourse, rendering that these frameworks fail to bridge the gap between ethical aspirations and operational accountability in agricultural data governance.

Yu et al. (2025) revealed that ATPs' data governance perspectives evolve significantly across development stages: early-stage startups, prioritizing market entry through simplified farmer relationships, often adopt open data ownership stances to attract users; as ATPs mature, they increasingly construct technological barriers and deploy legal resources to draft contracts favoring corporate interests, complicating ownership definitions; ultimately, when acquired or scaled into large entities, ATPs shift toward conservative data ownership views, implementing aggressive and restrictive terms to maximize proprietary value extraction from agricultural data.

The foundational logic underpinning agricultural data governance practices centers on monetization potential—shifting focus beyond ownership ("who owns the data?") to control over value extraction ("who profits from data utilization?"). Consequently, data derivatives (e.g., reports, analytical insights, software, models, equipment) demand parallel consideration alongside raw data in governance frameworks. This derivative-centric approach is already emerging in federal policies. The NSF Public Access Plan 2.0 mandates public sharing of research data through managed repositories while encouraging, though not requiring, open access to derivatives (National Science Foundation, 2023). Similarly, USDA's Open Gov Plan 4.0 strongly mandates open agricultural data but only suggests open-source goals for derivatives like software and models (Agriculture, 2016). This policy asymmetry reveals a critical governance gap:

while raw data transparency advances, derivative commercialization rights remain ambiguously allocated, enabling ATPs to capture downstream value without equitable profit-sharing mechanisms.

3.2 Map of current logics (only the portion reflected in existing documents)

To better illustrate the complex relationships and practices that influence data governance in agriculture, a logic map is developed (Fig. 2) by analyzing the solicitated 150 + contractual documents. This framework synthesizes prevailing practices by mapping ownership determinations for both raw data and derivative works across key governance pathways, color-coded to distinguish foundational approaches.

The spectrum progresses from left (fully open access) to right (highly proprietary models). At the open extreme (orange pathway), exclusively observed in government initiatives like NSF or USDA research projects (Agriculture, 2016; National Science Foundation, 2023), data will be shared to the public through certain authorized web-portals (e.g., USDA Ag Data Commons, NSF Public Access Repository (NSF-PAR)), while data derivatives face no licensing restrictions. Critically, these frameworks lack formal mechanisms to share profits with data contributors despite relying on farm-generated data.

ATPs dominate the proprietary end, treating data and derivatives as core competitive assets. Among the collected documents, 24% of contracts explicitly award data ownership to ATPs (green pathway), 76% acknowledge farmer ownership (red pathway), typically for subsets like operational data. ATP-owned data models employ two primary enforcement strategies, IP and trade secret. For IP, as seen in John Deere's user manuals, rights of repair/modification is prohibited by framing data access as a violation to their integral IP (Smith, 2023; Tomko, 2023). For trade secret, exemplified by Climate LLC's 2016 EULA, user IP claims are voided by dismissing data's economic value and controlled disclosure for trade secret. (Climate LLC, 2016)

For farmer-owned data (red pathway), derivative ownership remains undefined in contracts. However, ATPs inherently claim these rights (indicated by red dashed lines) since farmers rarely contribute intellectual input to derivative developments. This reflects an industry trend: ATPs like Climate LLC now recognize farmer data ownership in their latest EULA (Climate LLC., 2024), while retaining derivative control, a strategic shift responding to ethical critiques and industry agreements (Agridigital, 2018; American Farm Bureau Federation, 2016; Copa-Cogeca et al., 2018). It should be noticed that ownership delineations alone cannot capture usage complexities. Contracts embed tailored provisions for internal R&D, third-party data sharing, marketing applications, etc. These activity-specific terms create layered governance realities beyond what ownership pathways suggest.

4 Discussion

4.1 Current trend of data governance in agriculture

Agricultural data governance practices have gradually evolved over the past decade, as evidenced by two landmark cases: the "Right to Repair" movement targeting John Deere and Climate LLC's shifting EULAs for its FieldView platform.

The "Right to Repair" movement emerged in 2015–2016 when farmers discovered John Deere's licensing terms restricted independent repairs to protect IP, forcing reliance on authorized dealers (Wiens, 2015). High repair costs and procedural barriers led to a 2022 class-action lawsuit alleging monopolization of tractor repair markets ("Horning Farms v. Deere & Co.," 2022). The U.S. Department of Justice reinforced farmers' legal standing in a 2023 Statement of Interest (U.S. Department of Justice, 2023). Subsequently, in January 2023, John Deere signed a Memorandum of Understanding (MOU) with the American Farm Bureau Federation, promising limited access to repair tools (AFBF & John Deere, 2023). This reflects incremental policy adaptation toward farmer autonomy.

In another example, Climate LLC (acquired by Monsanto in 2013, later a Bayer subsidiary) demonstrates shifting data governance through revisions to its EULA:

• 2016 EULA: Asserted exclusive corporate ownership of all generated data and derivative works: "*As between Climate and you, we are the sole owner of the Climate Products and Generated Data... We do not grant you any rights or licenses under any of our technology or intellectual property rights...* " (Climate LLC, 2016)

• 2019 EULA: Differentiated data types. Farmers retained ownership of personal data and voluntarily submitted "Customer Farm Data," while data from FieldView systems constituted "Climate Generated Works" (owned by Climate LLC), including derivative analytics:

"We own any works we generate ('Climate Generated Works'), including data, tools, analyses, results... made available to you in or by the FieldView Services." (Climate LLC., 2019)

The EULA explicitly states that "*Aggregated or Anonymized Information is not considered Customer Farm Data.*" (Climate LLC., 2019), excluding farmers from ownership or profits derived from aggregated datasets.

• 2024 EULA: Expanded Climate LLC's license to use farmer data beyond service delivery to include R&D and product development:

"*You grant Us a non-exclusive license to access use, reproduce, display, modify, and prepare derivative works based on Your Customer Farm Data … for our internal operations and research and development purposes...*" (Climate LLC., 2024)

These cases illustrate a gradual formalization of farmer rights—repair autonomy for John Deere customers and nominal ownership of voluntarily submitted data for Climate users. However, ATPs still retain critical advantages. John Deere's concessions remain limited. The current efforts still need to be improved by providing practical guidance for solving equipment problems. Climate LLC and other similar

big ATPs hold broad licenses to exploit farmer data for derivative products and R&D. ATPs exclusively profit from aggregated, anonymized data and IP derived from farm data, while farmers hold no claims to resulting value. Thus, despite increased transparency and recognition of certain farmer rights, ATP governance frameworks systematically preserve corporate control over data monetization and technological innovation.

4.2 Value chain of data

The emergence of data-driven agricultural technologies, including smart sensors, AI analytics, and robotic systems, has catalyzed significant market growth while generating novel profit streams across the data value chain. Current governance frameworks, however, failed in addressing the inequitable distribution of value derived from downstream data products. As illustrated earlier, merely granting nominal ownership of raw data, such as Climate LLC's recognition of farmer ownership over "Customer Farm Data", proves insufficient rights to control derivative uses or share in resulting profits. This creates a structural imbalance where ATPs retain broad licenses to exploit farm data for commercial innovations, effectively enabling unilateral monetization of derived insights.

These governance gaps manifest differently across services. In machinery manufacturers contexts like John Deere's repair restrictions, farmers directly experience tangible control limitations, leading to legal action. Conversely, digital value chains in remote sensing and analytics (e.g., Climate LLC's FieldView) obscure exploitation pathways; farmers rarely perceive how their data fuels downstream R&D or proprietary products, complicating claims to value. Unique conditions may exist for different service categories, not limited to the listed in Table 1. Resolving this requires redefining ownership to encompass two critical dimensions: unrestricted rights to repair, modify, or transfer equipment/machine without functional disruption; formal recognition as co-contributors to derivative innovations, entitling farmers to audit trails and profit-sharing mechanisms.

This redefinition confronts a fundamental tension that as farmers grow increasingly aware of their data's value and demand greater control (Wiseman et al., 2019), ATPs leverage technical and contractual advantages to protect competitive interests (Yu et al., 2025). ATPs impose data robustness standards (e.g., scale, granularity) that marginalize smaller operators, embed aggressive usage rights in non-negotiable agreements. Compounding this imbalance, farmers typically lack the technical expertise, legal resources, and institutional leverage to audit complex data flows or challenge opaque governance terms. This power asymmetry, where ATPs control infrastructure, define data standards, and unilaterally determine permissible uses, creates a systemic disadvantage for farmers (Carbonell, 2016; Jakku et al., 2019; Wiseman et al., 2019).

Under current conditions, this conflict appears to be a deadlock, one that is unlikely to be resolved without external intervention. Government action, whether through court cases, legislation updates, or regulatory reform, is essential to balance the game.

4.3 Vision of future

4.3.1 Full map of logic

The "Right to Repair" movement, while not directly governing data practices, signals growing recognition of farmers that may catalyze legal support for more ethical policies. This trend suggests future governance frameworks could increasingly acknowledge farmers' claims to data-derived value. As illustrated in Fig. 3, recognizing farmers as data owners necessitates clarifying extra derivative ownership pathways while precisely defining ATPs' rights.

Under existing collaborations, derivative products/services emerging from farm data could follow two distinct ownership models: exclusive ATP ownership (red pathway) or farmer-ATPs co-ownership (blue pathway). The Climate LLC model exemplifies exclusive ATP control, where services or decision-making assistances are contractually supported by farmer data contributions. This mode would be straightforward and easy to manage if such collaborations are isolated in the scope of the service delivering. But, in the Climate LLC model, the usage of data was extended to further innovation R&D which produce other data-driven profits and may cause potential conflict of sharing these profits. Similar terms are broadly included in contracts or manuals of many other ATPs. To avoid future conflicts, it requires stringent conditions: 1) farmers’ leverage, Farm-sourced data, whether collected by ATP equipment or otherwise, must remain farmer-owned; 2) restricted ATP rights, ATP data usage must be strictly limited to delivering contracted services, excluding profit-generating activities like internal R&D or marketing. These constraints would mitigate conflicts over downstream commercialization. But it brings the technology challenge of monitoring and tracing data activities.

Co-ownership represents a critical ethical evolution for future governance, especially when data-derivative profits exist. As farmers gain recognition as data contributors, their rights must include profit participation when ATPs commercialize derivatives (e.g., through marketing or R&D). Practical implementation remains challenging because of 1) determining fair profit shares is complex without transparent data lineage tracking and 2) auditing derivative value remains technologically difficult. Alternative solutions include one-time data purchases, where ATPs compensate farmers through service discounts or upgrades for unlimited data rights, or adopting a "bill of data rights" framework (Berman & Hirschman, 2018; Tisne, 2018) regulating collection/usage without establishing property claims. The latter aligns with concepts like California's proposed 'data dividend'(Au-Yeung, 2019), requiring corporations to share revenue generated from user data. The one-time data purchase mode could be a short-term compromise for the current industry when technologies are not able to provide reliable data monitoring and auditing capabilities to support sharing long-term continuous profit generation from data resources.

Beyond primary ATP relationships, farmers should retain rights to whether monetize data externally or not. For instance, startups training AI models could purchase data from farmers or from markets where data is a new type of commodity, bypassing data ownership debates to engage directly at the derivative-determination stage (Fig. 3, red/blue pathways). Farmers might also anonymize and openly share data, creating public-domain derivatives (purple pathway), further diversifying governance approaches. But

still, as data can be reused repeatedly and transformed into various type of information, which may produce long-term, multiple streams of profits, the tracking of dynamically growing life cycle of data flow and activities is of particularly importance.

4.3.2 Implications

4.3.2.1 Contracting

Establishing robust data rights for farmers necessitates addressing critical contractual complexities, particularly precise data categorization and explicit articulation of ownership rights. Agricultural decision-making relies on multi-dimensional data, including voluntarily provided by farmers, collected with or without ATP technologies, and potentially containing sensitive personal details. While collaborations between farmers and ATPs intuitively rely on farm-sourced information, making clear differentiation of data is essential for delineating ownership.

The evolution of Climate LLC's EULA illustrates this progression. The 2016 version generically referenced farmers' actions, "*By uploading, inputting, transmitting, storing or otherwise making Data available to the Climate Products*" (Climate LLC, 2016), without differentiating personal data and other farming related information. By 2019, the EULA formally categorized farmer-provided information into Personal Data (subject to privacy regulations) and Customer Farm Data, defined as: "*you or your authorized users upload, input, transmit, store or otherwise make available to Climate information about your farm operation, such as your field locations, maps or boundaries, recorded environmental, weather or climatic conditions, agronomic practices, crop losses, crop yields, field images, field notes and other information relating to your farm operation*" (Climate LLC., 2019). Farmers retain unambiguous ownership of both categories, an ethically sound foundation.

The core conflict emerges in usage rights. While data categorization resolves privacy concerns, Climate LLC's 2024 EULA grants itself "a *non exclusive license*" to exploit Customer Farm Data broadly, while stating "*We own any works we generate ("Climate Generated Works")*". This framework effectively requires farmers to relinquish control and value share rights over derivatives, analogous to third parties profiting from your property without compensation, an inequity often obscured by knowledge gaps and contractual opacity.

To advance equitable governance, ATP contracts must explicitly delineate the rights inherent to farmer data ownership. This includes definitive control over downstream applications (e.g., R&D, marketing), co-ownership of derivative products, and entitlement to proportional profit shares. While data categorization mitigates ambiguity, particularly for personal information, the explicit statement of these rights may facilitate meaningful contract negotiation. This framework, in turn, holds significant potential to stimulate innovation within the sector by potentially lowering market entry barriers for smaller ATPs. Larger ATPs, benefiting from stable market shares and consistent revenue streams, prioritize IP control to fortify technological barriers. Consequently, without pressure threatening their market position, adopting novel technologies is not of vital importance. In contrast, smaller ATPs primarily compete

through technological differentiation to secure market access and drive growth. A demonstrated willingness to negotiate equitable data governance terms offers these smaller ATPs a strategic advantage: leveraging ethical practices as a competitive differentiator to enter the market. This dynamic could incentivize startup participation in the industry, accelerating technological innovations while fostering an ecosystem of ethically fair data governance at the same time. Furthermore, as successful smaller ATPs are typically acquired by larger entities, their established ethical governance frameworks hold the potential to reshape the perspectives and practices of large ATPs. This process may ultimately catalyze a systemic transformation towards a more equitable industry ecosystem.

While profit-sharing mechanisms and negotiated contracts represent forward-looking developments for equitable governance, the immediate market imperative is enhancing transparency in ATP data governance terms. This requires improvements in multiple aspects: updating stakeholder literacy (farmers, ATPs, legal professionals), establishing equitable regulatory frameworks, and promoting ethical paradigms (e.g., one-time data purchase). The most important task is interpreting contractual terms governing the complex nature of data ownership and attendant rights. This complexity brings significant drafting challenges for ATPs and legal professionals, as business structures dictate intricate data flows and corresponding rights (Yu et al., 2025). For example, multi-ATP collaborations require complicate layered agreements to allocate rights per participant, inevitably bringing opacity and conflict where overlaps of data flow or usage exist. Meanwhile, farmers face a critical deficit in accessible legal interpretation; the rapid evolution of data technologies outpaces legal adaptation, leaving professionals potentially ill-equipped to decipher latent risks, compounded by high service costs.

Natural Language Processing (NLP) and AI offer transformative potential for addressing these contractual complexities. Current legal applications include extracting contracting parties, dates, and obligations (Lippi et al., 2019); identifying service exceptions through linguistic pattern analysis (Gao et al., 2012); and flagging problematic lease terms (Leivaditi et al., 2020). Advanced models like Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019) enable classification of complex clauses such as auto-renewability terms (Elwany et al., 2019), though domain-specific adaptation remains essential. The Contract Understanding Atticus Dataset (CUAD) (Hendrycks et al., 2021) further advances legal review capabilities. Despite these developments, agricultural data governance remains an underdeveloped application area—likely due to historically low privacy prioritization and limited public scrutiny. As governance conflicts escalate, this technological gap demands urgent attention.

4.3.2.2 Business complexity

The replicable nature of agricultural data intensifies governance challenges, as Hicks (2023) highlights, "data is abundant" in that it can be easily copied and modified multiple times by different entities. This characteristic extends value chain lifecycles while creating complex interdependencies. These dynamics are amplified by multi-entity business structures in ATP ecosystems (Yu et al., 2025), exemplified through two models, collaborative model and outsourcing model.

In collaborative frameworks, ATP A (providing farm management software) and ATP B (supplying soil sensors) integrate their technologies to enable irrigation monitoring. While functioning as equal partners, ATP A's contract grants farmers ownership of field data and requires consent for downstream uses. Meanwhile, ATP B's terms claim proprietary rights over all sensor-generated data streams, reserving licensing decisions exclusively to ATP B. This creates fundamental conflict when the same irrigation data is claimed by both farmers (through ATP A's agreement) and ATP B.

In outsourcing models, ATP C (a major agribusiness) acquires ATP D (a drone imagery startup), rebranding its services under ATP C's platform. Pre-acquisition, ATP D's contracts granted perpetual data licenses to farmers. Post-acquisition, ATP C's new terms assert exclusive ownership over all derivative products created from farm data. This hierarchical structure creates legacy conflicts where farmers' existing rights under ATP D's agreements clash with ATP C's expansive claims.

These contractual collisions generate legal concerns. Farmers may simultaneously receive contradictory demands: ATP A requiring consent for data commercialization while ATP B exercises unilateral licensing rights. ATP C may restrict data usage that ATP D's legacy terms permitted. When multiple entities assert control over the same dataset, especially when farmers believe they retain ownership through one agreement while ATPs claim proprietary rights through others, enforcement becomes practically impossible.

Resolving these problems requires multi-tier governance. ATPs should disclose all collaborators (e.g., ATP C revealing ATP D's involvement) and data flow maps during farmer contracting. Confidential inter-ATP agreements must explicitly define ownership boundaries (e.g., distinguishing raw field data from aggregated analytics), establish profit-sharing mechanisms for derivative products, and embed mutual audit rights. Smaller ATPs like emerging sensor companies could differentiate themselves ethically, for instance, offering farmers 5% royalties on commercialized algorithms derived from their data. Without such measures, fragmented rights across opaque contractual terms will impair trust and obstacle agricultural innovation.

4.3.2.3 Government intervention

Agricultural data governance lags behind healthcare's more mature frameworks due to the absence of equivalent regulatory structures. While healthcare benefits from overarching policies like HIPAA that standardize data sharing and privacy protection, agricultural data, largely devoid of personal identifiers, lacks equivalent guiding principles. This regulatory absence has created fragmented governance practices where ATPs often prioritize corporate interests over farmer rights. Government intervention is therefore critical to establish agriculture-specific regulations with legal enforceability surpassing contractual agreements. Such policies must redefine data as an inherent attribute of its generator rather than mere IP governed by copyright law.

The complexity of ATP-farmer contracting, compounded by farmers' limited bargaining power, necessitates regulatory defaults to protect stakeholder interests. A foundational framework should

establish baseline rights when contracts remain ambiguous:

• Farmers automatically retain co-ownership claims to data derivatives if identified as data owners

• Revocation rights allowing farmers to withdraw data permissions when ATP practices conflict with their interests, triggering immediate cessation of data usage

Globally divergent legal traditions further inform solution design. U.S. and Western frameworks emphasize individual entitlements to facilitate market exchange (Hicks, 2023; Madison et al., 2009). Conversely, African and Asian customary systems often recognize communal or land-based rights. Hicks (2023) suggests that exploring data commons mechanisms within these legal systems could offer valuable insights for developing solutions. Under this framework, rights are user-based rather than absolute. In agriculture case, rights could be attribute to lands rather than transfer along with the movement of resource. For example, an ATP may be granted the right to collect data and provide analytical services for decision-making in a specific land by farmers but may not be entitled to monetize data or its derivatives, as these are part of the rights associated with this land. This approach, however, presents a significant challenge for contract design, as it requires clear delineation of each stakeholder's rights and necessitates corresponding auditing or supervisory mechanisms to ensure compliance and accountability.

Tenancy arrangements introduce additional complexity. With approximately 39% of U.S. farmland leased (USDA National Agricultural Statistics Service, 2022), disputes may arise between landowners and operators regarding data ownership (Mintert et al., 2022). Updated farm lease contracts must explicitly address: 1) Ownership of data generated on leased land, 2) Profit-sharing mechanisms for monetized data, and 3) Authorization protocols for ATP engagements. Without clear terms, tenant farmers risk exclusion from data value chains they actively sustain, while landowners face unresolved liability. This complexity cascades to ATP contracts, requiring multi-party agreements to define revenue distribution and usage rights across stakeholders.

4.3.2.4 Technology-traceability

Effective agricultural data governance faces a critical technology limit: the inherent difficulty in tracing and auditing data activities across complex value chains (Hicks, 2023). Unlike regulated sectors with established monitoring frameworks, agriculture lacks mechanisms to log data transformations (e.g., compilation, aggregation, or modification) that occur during analytical processing. This opacity creates two compounding failures:

1. **Unverifiable compliance**: Suspected contractual violations (e.g., unauthorized data commercialization) cannot be substantiated without immutable activity records.

2. **Unenforceable rights**: Farmers’ theoretical rights to derivative profits or usage control remain absent proof associated with source data.

Current technological solutions are inadequate. Blockchain enables traceability only in rigidly structured environments, such as cryptocurrency transactions relying on immutable ledgers (a dataset in a special structure) or smart contracts for predetermined transformations. Agricultural data must be adaptively recombined across services, otherwise its value cannot be embodied. Such fluidity does not meet the constraints of blockchain. For instance, soil data from Farm A may be aggregated with weather data from Farm B, normalized through proprietary algorithms, and monetized as predictive analytics, a multi-step process with no existing system reliably tracks.

Consequently, profit-sharing models remain theoretical. Farmers cannot evidence their data's contribution to commercialized derivatives, while supervisory entities lack audit trails to validate claims. This traceability gap fundamentally obstructs three pillars of ethical governance: Accountability for unauthorized data use, Transparency in value derivation, and Equity in benefit distribution. Resolving this requires innovations beyond transactional ledgers. Next-generation tracing systems must 1) capture granular transformation events, 2) preserve lineage across format conversions and analytical workflows, and 3) interface with diverse ATP platforms without disrupting operational flexibility. Until such capabilities exist, contractual rights to profit participation will remain unenforceable, despite evolving policy frameworks.

5 Conclusion

This study elucidates the systemic inequities embedded in agricultural data governance, where contractual practices prioritize ATP control over data derivatives while granting limited rights to farmer ownership of raw data. Analysis of ATP contracts and policy documents confirms that intellectual property (IP) frameworks and trade secret claims cause legal ambiguities of data ownership, enabling unilateral value extraction from farm-generated data. Although ethical initiatives (e.g., the EU Code of Conduct) affirm farmer rights, their non-binding nature and the rapid technological evolution outpacing legal adaptation render them ineffective. The derivative ownership gap, where farmers hold no claims to profits from data-driven innovations, remains the core injustice, compounded by ATPs' strategic shift toward restrictive governance as they scale. To address this imbalance, regulatory defaults must establish farmer co-ownership of derivatives, enforceable profit-sharing mechanisms, and rights protection protocols. Simultaneously, technological innovations in data lineage tracking are imperative to audit value chains and ensure accountability. Future efforts should prioritize multi-tier governance frameworks that harmonize stakeholder incentives, adapt to communal and land-based rights traditions, and incentivize ethical startups. Without such interventions, the agricultural sector risks entrenching data monopolies that undermine both equity and sustainable innovation.

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5.4 Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT (GPT-5) in order to improve the language and check gramma issues only. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Author Contribution

Ziwen Yu conducted the primary investigation, prepared datasets and reports, and drafted the manuscript. Songzi Wu contributed to data analysis, provided critical feedback, and reviewed and edited the manuscript. Both authors read and approved the final version of the paper.

Data Availability

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Figures

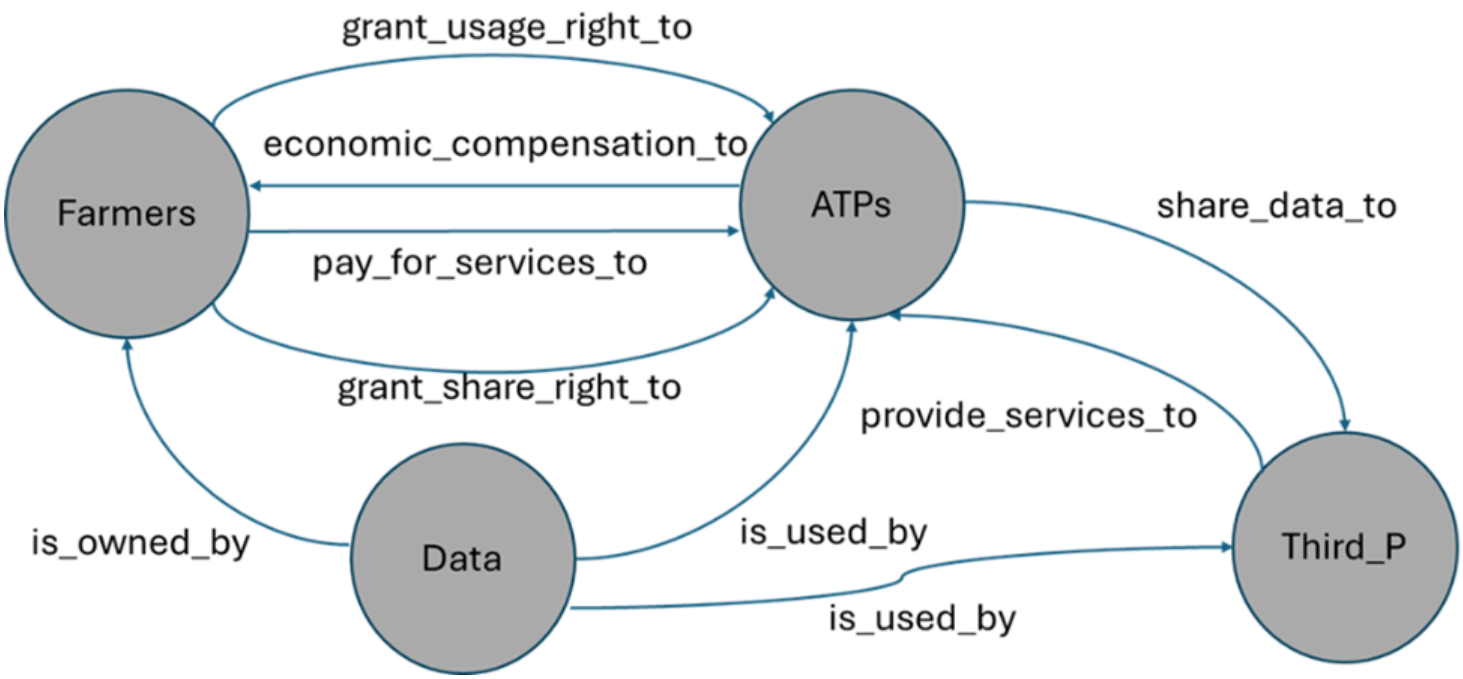

Figure 1

Sample knowledge graph illustrating the relationship representation of underlying logics in contracts

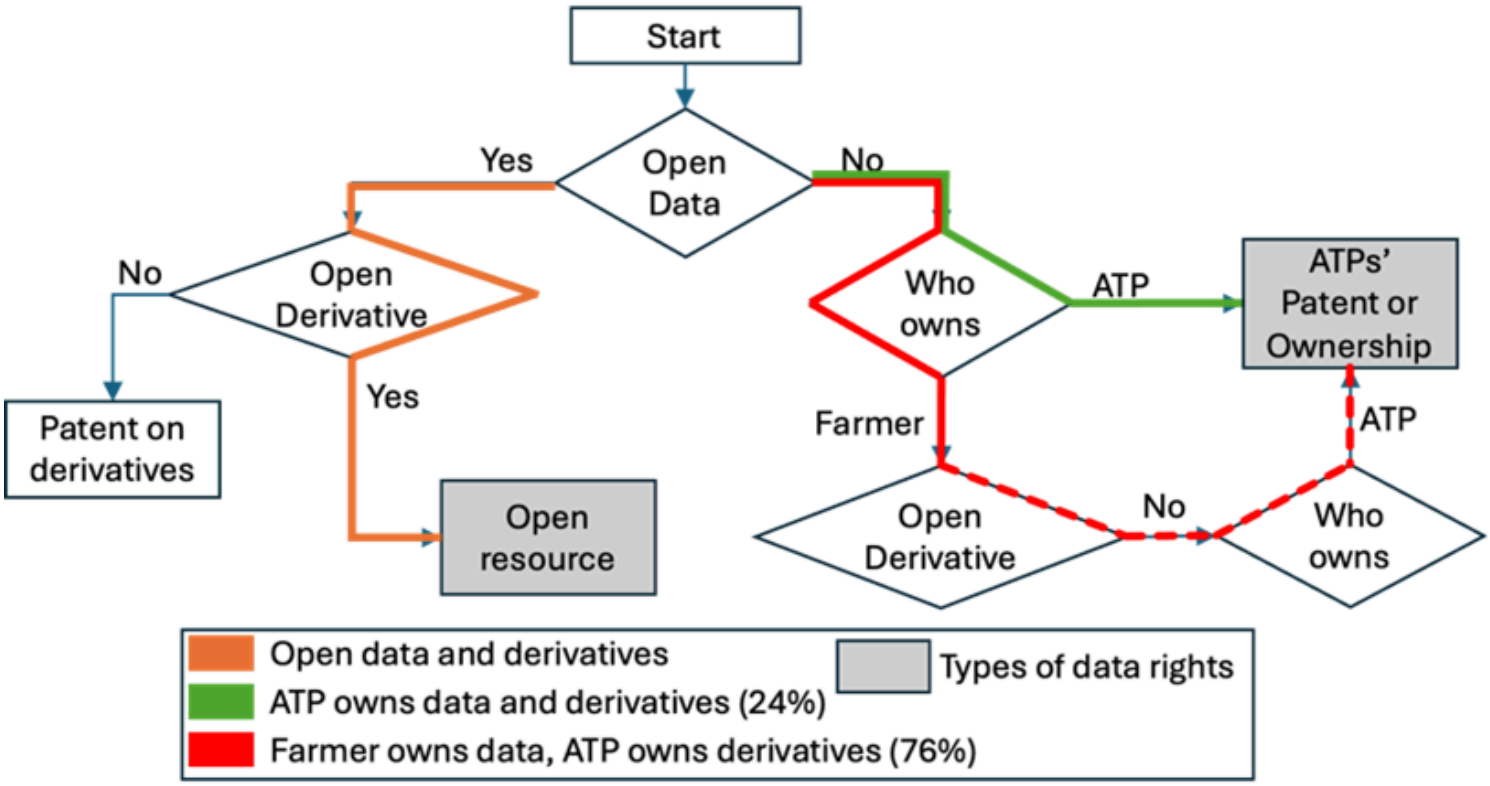

Figure 2

Map of data governance logics extracted from collected documents. Percentage reflects the portion of collected contractual documents that follow the corresponding pathway.

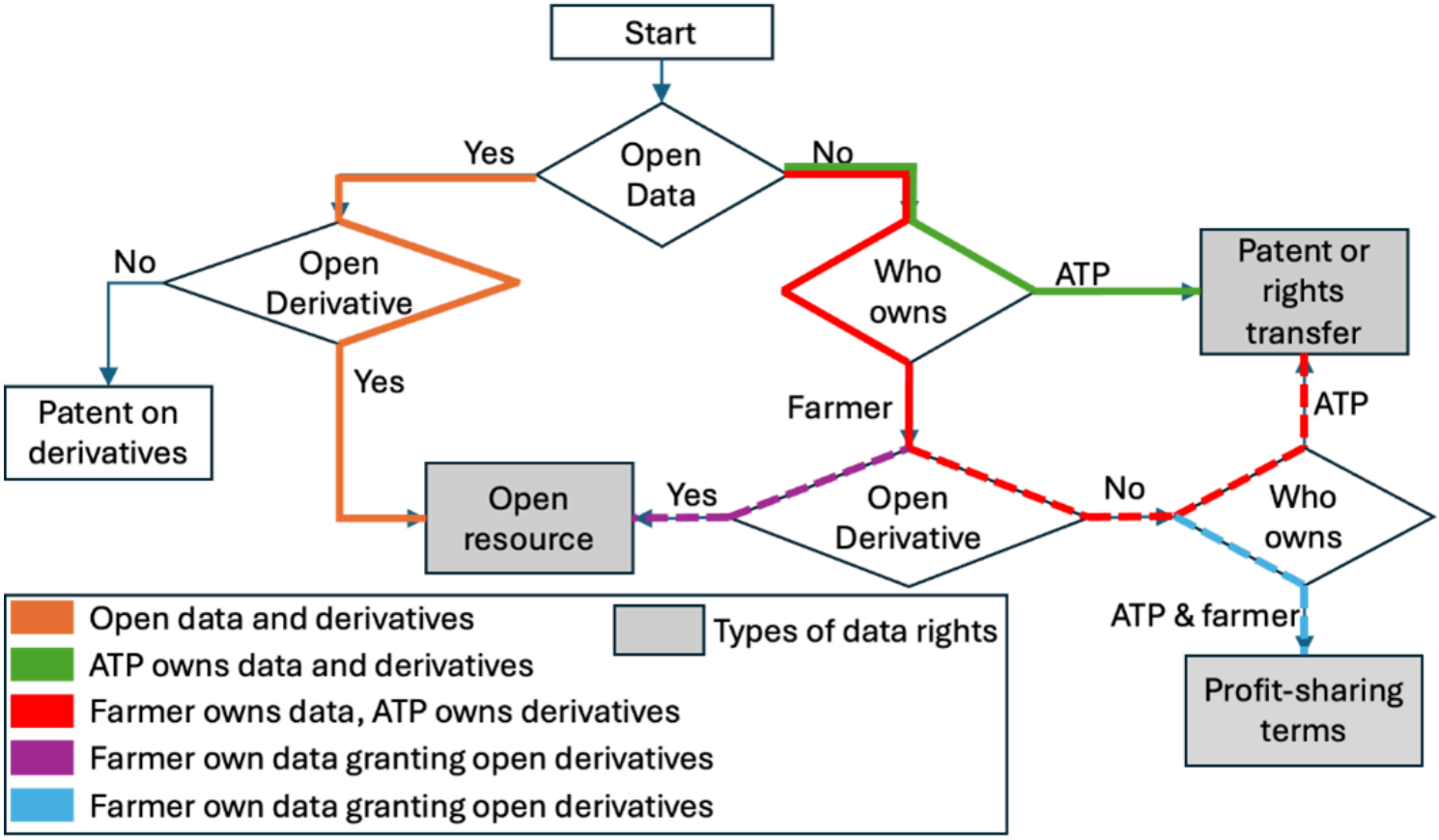

Figure 3

Map of data governance logics projected for future agriculture systems